

CANoe - ASK DLL 사용법

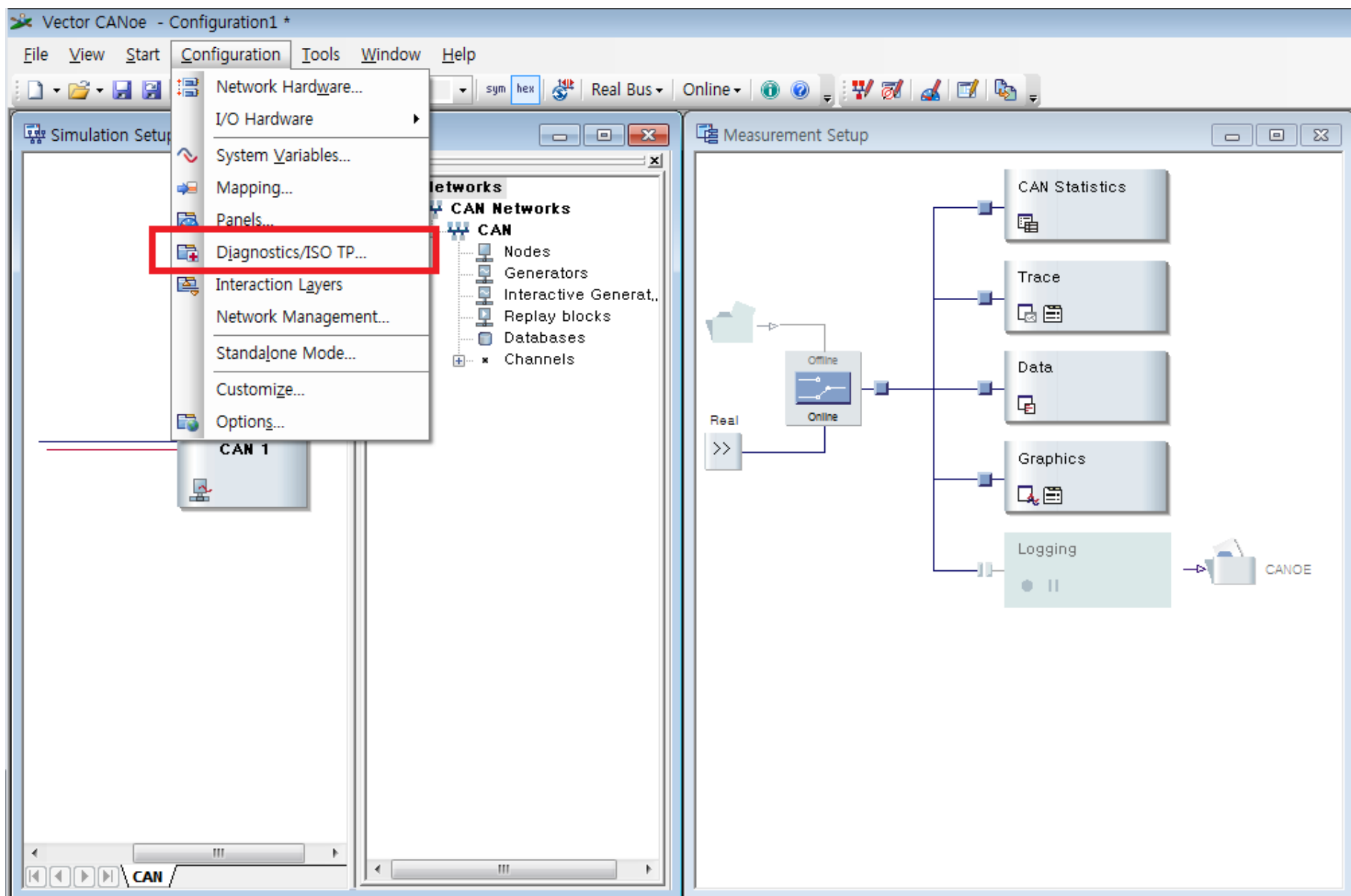
현대오토에버 차량보안기술팀

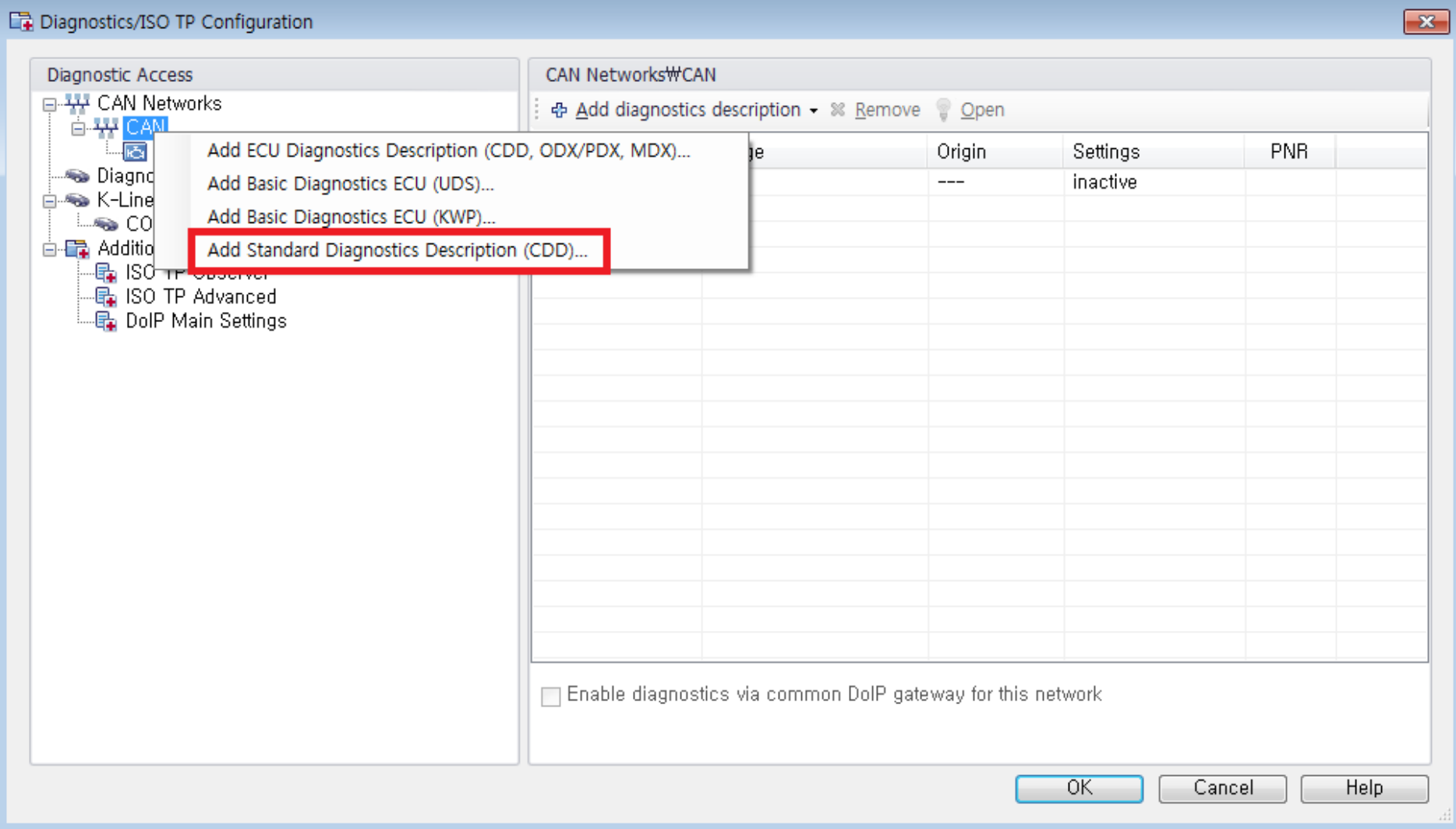
2017년 3월

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순서1. Seedkey DLL 파일 경로 설정







- CAN Networks
 - CAN
 - OBD-II Functionality
- Diagnostic over IP
- K-Line
- COM
- Additional Diagnostic Settings
 - ISO TP Observer
 - ISO TP Advanced
 - DoIP Main Settings

[Add diagnostic description](#) [Remove](#) [Open](#)

[illegible]

KWP 일 경우

OK Cancel Help

pay for this network

Help

Diagnostics/ISO TP Configuration

Diagnostic Access

- CAN Networks
 - CAN
 - SEEDKEY_TEST
 - Physical Network Request
 - Diagnostic Layer
 - Security DLL
 - OBD-II Functionality
 - Diagnostic over IP
 - K-Line
 - COM
 - Additional Diagnostic Settings
 - ISO TP Observer
 - ISO TP Advanced
 - DolP Main Settings

CAN Networks\CAN\SEEDKEY_TEST

ECU qualifier

SEEDKEY_TEST

노드 이름 설정

File

C:\Program Files (x86)\Vector CANoe 8.2\Exec32\Sta...GenericUDS.cdd

Available options

Variant

Common

Interface

Normal 11Bit

Language

en-US

Hint: The language selection does not affect the qualifiers used in CAPL programs.

Target group

(Display all services)

Hint: Choosing a target group restricts the set of services offered in the diagnostics console.

Usage of the diagnostics description

- ☐ Diagnostics tester ("send-only")...
- ☒ Physical Network Requests...
- ☐ Functional Group Requests...
- ☐ Assign to database node...
- ☐ Interpretation Only...

OK

Cancel

Help

Diagnostics/ISO TP Configuration

Diagnostic Access

- CAN Networks
 - CAN
 - SEEDKEY_TEST
 - Transport Layer
 - Diagnostic Layer
 - Security DLL**
 - ODD-II Functionality
- Diagnostic over IP
- K-Line
- COM
- Additional Diagnostic Settings
 - ISO TP Observer
 - ISO TP Advanced
 - DolP Main Settings

CAN Networks\CAN\SEEDKEY_TEST\Security DLL

This page is used to set the Seed&Key DLL. The DLL is needed when using the CAPL function 'DiagGenerateKeyFromSeed'.

Security DLL



Seedkey DLL 경로 입력

OK

Cancel

Help

순서2. CAPL에서 DiagGenerateKeyFromSeed() 함수 실행

- 이후 CAPL Browser 에서
diagSetTarget(“노드이름”) 호출 후,
DiagGenerateKeyFromSeed() API를 호출하면 경로 설정한 DLL 파일이 실행됩니다.
- DiagGenerateKeyFromSeed()의 자세한 API 설명 및 예제 코드는 CAPL Browser의 HELP에서 DiagGenerateKeyFromSeed() API를 검색하시기 바랍니다.

DiagGenerateKeyFromSeed() API 설명

diagGenerateKeyFromSeed

CAPL Function Overview » Diagnostics » diagGenerateKeyFromSeed

Syntax	<code>Long diagGenerateKeyFromSeed (byte seedArray[], dword seedArraySize, dword securityLevel, char variant[], char ipOption[], byte keyArray[], dword maxKeyArraySize, dword& keyActualSize)</code>
Function	Creates a key to execute secured diagnostic functions within devices. The key will be defined with the Seed of the device.
Parameters	seedArray Seed for the definition of the key.
	seedArraySize Number of bytes in the SeedArray.
	securityLevel Security level for which the key will be created.
	variant Variant of the diagnostic description.
	ipOption Further options that will be forwarded to the customer function.
	keyArray Key that is created with the customer DLL.
	maxKeyArraySize The maximum allowed number of bytes in the keyArray.
	keyActualSize Actual used bytes in the keyArray.
Return Values	At success 0 will be returned, otherwise an error code will be returned. For further error analyses you can use the callback function <code>_diag_GetError</code> .

DiagGenerateKeyFromSeed() 예제 코드

```
on start
{
    if( 0 != diagSetTarget("SEEDKEY_TEST")) write( "Error setting target!");
    setTimer(ASK_TEST, 500);
}

on ASK_TEST
{
    // ----- (필수)SeedKey 변수 설정 -----
    byte seedArray[8];
    int  seedArraySize      = 8;
    int  securityLevel      = 0x01;
    char variant[200]       = "Common";
    char option[200]        = "";
    byte keyArray[8];
    int  maxKeyArraySize    = 8;
    DWORD actualSize        = 0;

    // seed를 seedArray 배열에 넣어준다.
    ...
    // 결과값은 keyArray에 들어온다.
    DiagGenerateKeyFromSeed (seedArray, seedArraySize, securityLevel, variant,
    option,                  keyArray, maxKeyArraySize, actualSize);
}
```

부록. 예제 코드 테스트 결과 (64bit CANoe)

CANoe 9.0x, 64bit호환

